Colab Notes

Some Positive properties

1. Zero setup / instant environment

- No local installs, no dependency hell

2. Free (and paid) GPU / TPU access

- Free tier gives limited GPU
- Deep learning
- LSTM / Transformers
- Large matrix ops

3. Cloud-based & portable

- Runs entirely in browser
- Works on **any machine** (Mac, Windows, Linux, low-spec laptops)

4. Preinstalled ML stack – for quick prototyping

- NumPy, pandas, matplotlib
- scikit-learn
- PyTorch, TensorFlow
- XGBoost, LightGBM

Colab Notes

Some challenging aspects

- Sessions reset unexpectedly (long pymc, long MCMC…)
- You cannot pin this image – package version control a must beyond core auto installed packages

```
pip install \
        numpy==1.26.4 \
        scipy==1.11.4 \
        pymc==5.10.0 \
        arviz==0.17.0
Etc.
```

- I/O is slow compared to local SSDs

- Specific caution wrt. confidence in results without re-running locally

```
from google.colab import drive
drive.mount('/content/drive/')

filePath = "/content/drive/My Drive/data/keyDevs_out_NVDA_Rel_LLM_Exploded.csv"
df = pd.read_csv(filePath)
df.head()
```

Colab, compressive example

This takes a while to install once you connect to Colab instance

```
!pip install git+https://github.com/kieferk/pymssa.git
!pip install git+https://github.com/aaren/wavelets
!pip install fredapi
!pip install pykalman
!pip install pandas_datareader
!pip install dfencoder
!pip install neurodsp
!pip install dcor
!pip install cif
!pip install bycycle
!pip install ortools
!pip install tigramite
!pip install stumpy
!pip install -U matplotlib
!pip install causal_ccm
!pip install git+https://github.com/majianthu/pycopent.git
!pip install PyWavelets
!git clone https://github.com/bmfreis/recurrence_python.git
%cd recurrence_python
!pip install ssqueezepy
```